

Practical 14 - Models for rates

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `rubber.csv`

The purpose of this exercise is to fit and interpret some Poisson regression models for data from a cohort study of men from the rubber manufacturing industry.

In one rubber factory isolated conditions were maintained for the manufacturing process, while in a second workers were exposed to more dirt and fumes. The aim is to investigate differences in death rates between men who worked in the two factories, making appropriate allowances for age.

The dataset (`rubber.csv`) consists of four variables for groups defined by factory and age category:

Variable	Description
<code>agegrp</code>	Age group: 1 = 50–59, 2 = 60–69, 3 = 70–79, 4 = 80–89
<code>factory</code>	Factory (1 or 2)
<code>deaths</code>	Number of deaths
<code>pyrs</code>	Number of man-years of exposure

Aims

The aim of this session is to address the following questions about these data:

1. Is there a difference between deaths rates in the two factories?
2. Is death rate associated with age?
3. What is the best way to model age (categorical or continuous)?

Analyses

Question 1

Calculate the total number of deaths and total person years of observation. What is the overall rate of death amongst these men? What is the log death rate?

SOLUTIONS

```
rubber <- read.csv("rubber.csv")

rubber$agegrp <- factor(rubber$agegrp,
                        levels = c("1", "2", "3", "4"),
                        labels = c("50-59", "60-69", "70-79", "80-89"))

sum(rubber$deaths)
```

```
[1] 160
```

```
sum(rubber$pyrs)
```

```
[1] 19345
```

```
#rate
sum(rubber$deaths)/sum(rubber$pyrs)
```

```
[1] 0.008270871
```

```
#log rate
log(sum(rubber$deaths)/sum(rubber$pyrs))
```

```
[1] -4.795015
```

The death rate is better expressed as 8.27 deaths per 1000 years of follow-up.

Question 2

Calculate the death rate and log death rate in each factory.

SOLUTIONS

You could of course do the following using some base R commands, but here we use some dplyr package functionality:

```
library(dplyr)
rubber %>%
  group_by(factory) %>%
  summarise(deaths=sum(deaths), pyrs=sum(pyrs),
            death_rate = sum(deaths)/sum(pyrs),
            log_death_rate = sum(deaths)/sum(pyrs))
```

```
# A tibble: 2 x 5
  factory deaths  pyrs death_rate log_death_rate
  <int>   <int> <int>      <dbl>        <dbl>
1     1     72  9774     0.00737      0.00737
2     2     88  9571     0.00919      0.00919
```

In factory 1: Death rate = $72/9774 = 0.00737$ deaths per year (7.37 deaths per 1000 years of follow-up). Log death rate = -4.941 .

In factory 2: Death rate = $88/9571 = 0.00919$ deaths per year (9.19 deaths per 1000 years of follow-up). Log death rate = -4.689 .

Question 3

Generate a variable `log_death_rate` in the data frame that calculates the log death rates in each row.

Next, complete the following `ggplot2` code to plot the log death rates against age group, with points labelled according to factory. What does this plot suggest?

```
library(ggplot2)
rubber %>% ggplot(aes(agegrp, log_death_rate)) +
  geom_point(size = 3) +
  geom_text(aes(label = ), position = position_nudge(x = 0.1)) +
  labs(x = "Age group", y = "Log death rate")
```

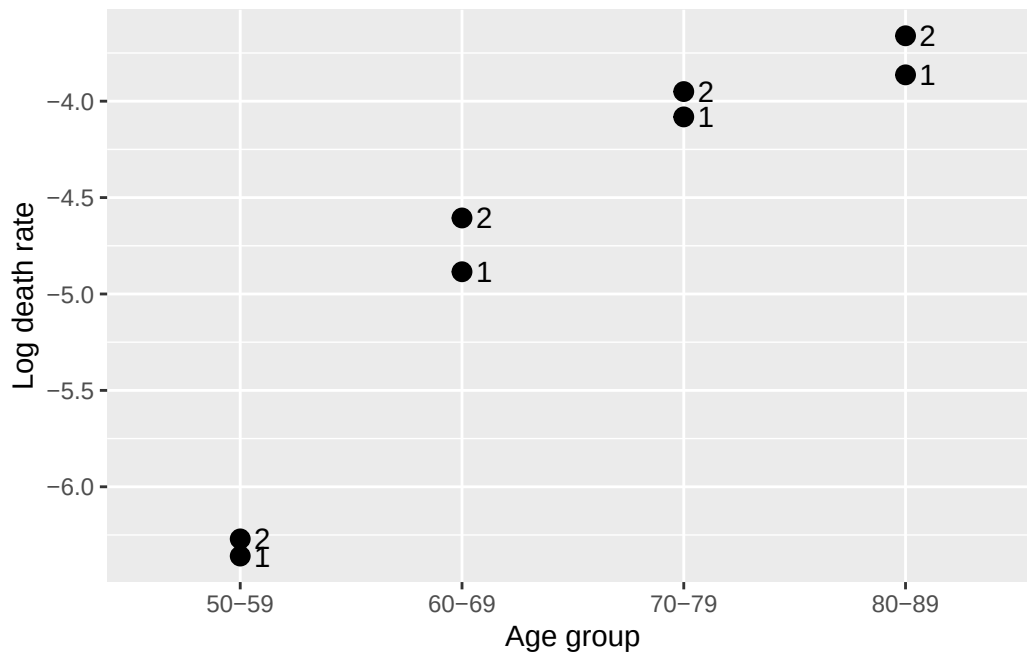
SOLUTIONS

```

rubber$log_death_rate = log(rubber$deaths/rubber$pyrs)

library(ggplot2)
rubber %>% ggplot(aes(agegrp, log_death_rate)) +
  geom_point(size = 3) +
  geom_text(aes(label = factory), position = position_nudge(x = 0.1)) +
  labs(x = "Age group", y = "Log death rate")

```



From this plot we can see that rates increase with age in both factories, and that age-specific rates are higher in factory 2 for all age groups. The age trend shows sign of flattening off at the oldest age category, suggesting some non-linearity in that relationship.

Question 4

Write down algebraically a model that can be used to investigate the relationship between death rate and age group (treated as a categorical variable, with age group 50-59 taken as the baseline category).

Take a moment to check with your colleagues that you have specified the models in the same way.

SOLUTIONS

Let Y_i be the number of deaths in group i ($i = 1, \dots, 8$) with exposure time t_i years. We assume that the Y_i are independent, with

$$Y_i \sim \text{Po}(\mu_i),$$

where

$$\mu_i = \lambda_i t_i.$$

Here λ_i is the underlying mortality rate in group i .

The link function and linear predictor are given by

$$\eta_i = \log(\mu_i) = \log(t_i) + \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3}.$$

The covariates are defined as follows:

$$x_{i1} = \begin{cases} 0 & \text{if the } i\text{th group is not aged 60--69} \\ 1 & \text{if the } i\text{th group is aged 60--69} \end{cases}$$

$$x_{i2} = \begin{cases} 0 & \text{if the } i\text{th group is not aged 70--79} \\ 1 & \text{if the } i\text{th group is aged 70--79} \end{cases}$$

$$x_{i3} = \begin{cases} 0 & \text{if the } i\text{th group is not aged 80--89} \\ 1 & \text{if the } i\text{th group is aged 80--89} \end{cases}$$

Question 5

Using R:

- (a) obtain the maximum likelihood estimates of the parameters in your model. How strong is the evidence for an age effect?

SOLUTIONS

```
# Create log person-years var for offset
rubber$lpyrs <- log(rubber$pyrs)

# Fit model without age
poisInt <- glm(deaths ~ offset(lpyrs), family = poisson, data = rubber)
summary(poisInt)
```

```
Call:
glm(formula = deaths ~ offset(lpyrs), family = poisson, data = rubber)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.79502	0.07906	-60.65	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 103.88 on 7 degrees of freedom
Residual deviance: 103.88 on 7 degrees of freedom
AIC: 142.73

Number of Fisher Scoring iterations: 5

```
# Fit a model with age
poisAge <- glm(deaths ~ agegrp + offset(lpyrs), family = poisson, data = rubber)
summary(poisAge)
```

```
Call:
glm(formula = deaths ~ agegrp + offset(lpyrs), family = poisson,
    data = rubber)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-6.3159	0.2673	-23.632	< 2e-16 ***
agegrp60-69	1.5828	0.2950	5.365	8.11e-08 ***
agegrp70-79	2.3030	0.2946	7.816	5.45e-15 ***
agegrp80-89	2.5547	0.3609	7.079	1.46e-12 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 103.8826 on 7 degrees of freedom
Residual deviance: 1.7108 on 4 degrees of freedom
AIC: 46.553

Number of Fisher Scoring iterations: 4

```
# Compare the models
anova(poisInt, poisAge)
```

Analysis of Deviance Table

```
Model 1: deaths ~ offset(lpyrs)
Model 2: deaths ~ agegrp + offset(lpyrs)
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1         7    103.883
2         4     102.17  3    102.17 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The test statistic here is the difference in the log likelihoods multiplied by minus 2 ($102.17 = -2 \times (-70.363 - (-19.277))$). This test gives overwhelming evidence for an effect of age.

(b) calculate the estimated rate ratios comparing age groups:

- i) 60-69 with 50-59
- ii) 80-89 with 70-79

For i) calculate a confidence interval for the rate ratio.

Optional additional exercise: calculate a standard error for ii) using the values stored in `vcov(yourmod)`, and then a CI based on this.

SOLUTIONS

```
#rate ratio comparing 60-69 with 50-59 (reference level) is
exp(coef(poisAge)[2])
```

```
agegrp60-69
4.868732
```

```
#SE from model output for this coef is 0.2950
exp(coef(poisAge)[2]-1.96*0.2950)
```

```
agegrp60-69
2.730906
```

```
exp(coef(poisAge)[2]+1.96*0.2950)
```

```
agegrp60-69  
8.680104
```

```
#second log rate ratio of interest is  
iiEst <- coef(poisAge)[4]-coef(poisAge)[3]  
#so rate ratio comparing 80-89 with 70-79 is  
exp(iiEst)
```

```
agegrp80-89  
1.286225
```

```
#variance of this: use Var(A-B)=Var(A)+Var(B)-2*Cov(A,B)  
vcov(poisAge)
```

```
              (Intercept) agegrp60-69 agegrp70-79 agegrp80-89  
(Intercept)  0.07142857 -0.07142857 -0.07142857 -0.07142857  
agegrp60-69 -0.07142857  0.08705357  0.07142857  0.07142857  
agegrp70-79 -0.07142857  0.07142857  0.08681319  0.07142857  
agegrp80-89 -0.07142857  0.07142857  0.07142857  0.13025210
```

```
vcov(poisAge)[4,4] + vcov(poisAge)[3,3] - 2*vcov(poisAge)[3,4]
```

```
[1] 0.07420814
```

```
#SE is then sqrt of this  
iiSE <- sqrt(vcov(poisAge)[4,4] + vcov(poisAge)[3,3] - 2*vcov(poisAge)[3,4])  
#and finally CI  
exp(iiEst-1.96*iiSE)
```

```
agegrp80-89  
0.7541115
```

```
exp(iiEst+1.96*iiSE)
```

```
agegrp80-89  
2.193808
```

```
#note you could also get the marginalEffects package to do this
```

Question 6

Now include both age group and factory effects as covariates. What is the evidence for a difference in death rates between the factories (conditional on age group)? Calculate the estimate (and 95% CI) of the rate ratio comparing the two factories, conditional on age group.

SOLUTIONS

```
poisAgeFact <- glm(deaths ~ agegrp + factor(factory)+
  offset(lpyrs),
  family = poisson, data = rubber)
summary(poisAgeFact)
```

Call:

```
glm(formula = deaths ~ agegrp + factor(factory) + offset(lpyrs),
    family = poisson, data = rubber)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-6.4132	0.2800	-22.901	< 2e-16 ***
agegrp60-69	1.5768	0.2951	5.343	9.12e-08 ***
agegrp70-79	2.2976	0.2947	7.797	6.34e-15 ***
agegrp80-89	2.5545	0.3609	7.078	1.46e-12 ***
factor(factory)2	0.1939	0.1590	1.220	0.223

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 103.88261 on 7 degrees of freedom
Residual deviance: 0.21568 on 3 degrees of freedom
AIC: 47.058

Number of Fisher Scoring iterations: 3

```
anova(poisAge,poisAgeFact)
```

Analysis of Deviance Table

```
Model 1: deaths ~ agegrp + offset(lpyrs)
Model 2: deaths ~ agegrp + factor(factory) + offset(lpyrs)
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1         4    1.71077
2         3    0.21568  1    1.4951  0.2214
```

```
#manual calc of the p-value
pchisq(1.4951,df=1,lower.tail = FALSE)
```

```
[1] 0.2214268
```

Note the similarity between the p-values from the Wald test of the factory effect ($z = 1.220$, hence the equivalent χ_1^2 test statistic is $1.220^2 = 1.489$, $p = 0.223$) and the likelihood ratio test given here. These tests are asymptotically equivalent.

```
exp(0.1939)
```

```
[1] 1.213975
```

```
exp(0.1939-1.96*0.1590)
```

```
[1] 0.8889271
```

```
exp(0.1939+1.96*0.1590)
```

```
[1] 1.657881
```

Conditional on age group, the death rate for factory 2 is 21% higher than that for factory 1 is (rate ratio = 1.21, 95% confidence interval (0.89, 1.66)), but this is not statistically significant.

Question 7

Now fit a model that includes the interaction between age group and factory.

- (a) What term can be used to describe this model (Hint: look at its deviance)?
- (b) What is the evidence for an interaction? How do you interpret this for these data?
- (c) Write down algebraically the linear predictor for this model, and use this to help you calculate the estimated rates of death in
 - i) factory 1, age 70-79
 - ii) factory 2, age 50-59
 - iii) factory 2, age 60-69
- (d) Check your answers against your original data

Take a moment to check with your colleagues that you have specified the models in the same way.

SOLUTIONS

```
poisAgeFactInter <- glm(deaths ~ agegrp*factor(factory)+
                        offset(lpyrs),
                        family = poisson, data = rubber)
summary(poisAgeFactInter)
```

Call:

```
glm(formula = deaths ~ agegrp * factor(factory) + offset(lpyrs),
     family = poisson, data = rubber)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-6.35933	0.37796	-16.825	< 2e-16 ***
agegrp60-69	1.47456	0.42414	3.477	0.000508 ***
agegrp70-79	2.27784	0.41975	5.427	5.74e-08 ***
agegrp80-89	2.49597	0.51755	4.823	1.42e-06 ***
factor(factory)2	0.08888	0.53452	0.166	0.867939
agegrp60-69:factor(factory)2	0.19018	0.59142	0.322	0.747789
agegrp70-79:factor(factory)2	0.04246	0.58959	0.072	0.942587
agegrp80-89:factor(factory)2	0.11377	0.72237	0.157	0.874854

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

```

Null deviance: 1.0388e+02 on 7 degrees of freedom
Residual deviance: 1.7097e-14 on 0 degrees of freedom
AIC: 52.843

```

Number of Fisher Scoring iterations: 3

- (a) This is a saturated model, because the number of parameters is the same as the number of (grouped) observations. Because of this the deviance for this model is 0.

```
anova(poisAgeFact,poisAgeFactInter)
```

Analysis of Deviance Table

Model 1: deaths ~ agegrp + factor(factory) + offset(lpyrs)

Model 2: deaths ~ agegrp * factor(factory) + offset(lpyrs)

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	3	0.21568			
2	0	0.00000	3	0.21568	0.975

- (b) The likelihood ratio test comparing the fit of the saturated model (which includes the interaction term) with the model in part 6 (which does not) provides no evidence of an age-by-factory interaction ($p = 0.975$). Note that the test statistic here (0.22) is identical to the residual deviance for the model without the interaction in part 6 so strictly we did not need to fit this model in order to test for evidence of an interaction.
- (c) The linear predictor is given by

$$\eta_i = \log(t_i) + \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 z_i + \beta_5 (x_{i1} z_i) + \beta_6 (x_{i2} z_i) + \beta_7 (x_{i3} z_i).$$

The covariates are defined as follows:

$$x_{i1} = \begin{cases} 0 & \text{if the } i\text{th group is not aged 60--69} \\ 1 & \text{if the } i\text{th group is aged 60--69} \end{cases}$$

$$x_{i2} = \begin{cases} 0 & \text{if the } i\text{th group is not aged 70--79} \\ 1 & \text{if the } i\text{th group is aged 70--79} \end{cases}$$

$$x_{i3} = \begin{cases} 0 & \text{if the } i\text{th group is not aged 80--89} \\ 1 & \text{if the } i\text{th group is aged 80--89} \end{cases}$$

$$z_i = \begin{cases} 0 & \text{if the } i\text{th group is from factory 1} \\ 1 & \text{if the } i\text{th group is from factory 2} \end{cases}$$

For men in factory 1 aged 70–79, the estimated rate of death is

$$\exp(\hat{\beta}_0 + \hat{\beta}_2) = \exp(-6.359 + 2.278) = 16.9 \text{ per 1000 person-years.}$$

For men in factory 2 aged 50–59, the estimated rate of death is

$$\exp(\hat{\beta}_0 + \hat{\beta}_4) = \exp(-6.359 + 0.089) = 1.89 \text{ per 1000 person-years.}$$

For men in factory 2 aged 60–69, the estimated rate of death is

$$\exp(\hat{\beta}_0 + \hat{\beta}_1 + \hat{\beta}_4 + \hat{\beta}_5) = \exp(-6.359 + 1.475 + 0.089 + 0.190) = 10.0 \text{ per 1000 person-years.}$$

(d) These answers agree with the rates in the raw data.

Question 8

Now consider the age group variable as continuous. Fit a model with age group and factory as covariates and consider its residual deviance.

What do you conclude about the overall fit of the model with continuous age and factory as additive effects? How might the fit be improved?

SOLUTIONS

To fit the model with `agegrp` treated as continuous we will first need to convert it back to a numeric.

```
rubber$agegrp <- as.numeric(rubber$agegrp)
poisAgeCtsFact <- glm(deaths ~ agegrp + factor(factory)+
                      offset(lpyrs),
                      family = poisson, data = rubber)
summary(poisAgeCtsFact)
```

```
Call:
glm(formula = deaths ~ agegrp + factor(factory) + offset(lpyrs),
    family = poisson, data = rubber)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-6.66597	0.24366	-27.358	<2e-16 ***
agegrp	0.80643	0.08422	9.575	<2e-16 ***
factor(factory)2	0.20836	0.15891	1.311	0.19

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 103.883 on 7 degrees of freedom
Residual deviance: 12.712 on 5 degrees of freedom
AIC: 55.555

Number of Fisher Scoring iterations: 4

The residual deviance is 12.712. Under the null hypothesis that the model is correct, this test statistic follows a chi-squared distribution on 5 degrees of freedom. The p-value can be calculated as:

```
pchisq(12.712,df=5,lower.tail = FALSE)
```

```
[1] 0.02623243
```

We thus have some evidence against the model being correct. It is probably best to revert to models which treat age as categorical.

Alternatively, one could try adding a quadratic term, but the coefficients in the categorical model are easier to interpret. Given that there are only 3 age-parameters in this model (only one less than in the quadratic model) the categorical model is probably the best choice.

Question 9

Now consider your analysis as a whole. What are your epidemiological conclusions from this analysis? Working together with one or more colleagues, write a short paragraph to summarise your findings concerning the comparison of the two factories.

SOLUTIONS

In conclusion there is strong evidence ($p < 0.001$) that death rates increase with age. The rate in 60-69 year olds was 4.8 (95% CI 2.7 to 8.6) times that in 50-59 year olds. Thereafter the rate increased in each decade but the relative decade-on-decade increases were smaller.

Death rates were 21% higher (95% CI 11% lower to 66% higher) in the second factory than in the first conditional on age group, but this difference was not statistically significant ($p = 0.22$). There was no evidence that the rate ratio comparing the two factories differed according to age.